

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: **Trace (solo team)**

Project / Product Name: **PRISM Code Search: CPU-friendly code retrieval (Theme 1, Agentic Code Intelligence)**

Organization / Institution (if any): **Thapar Institute of Engineering and Technology, Patiala**

Submission Date: 30th September 2026

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project?
Yes

Yes. Claude was used. Through Claude Code it helped write implementation generic code, and tests as per my instructions. The evaluations ran on my own Kaggle, Colab and Windows PC accounts.

3. Purpose of AI Usage (Brief Details)

Idea/brainstorming: The plan and the three differentiator ideas (query-to-code HyDE, multi-view snippets, stable-core + version-delta) came from a Build Plan document that I gave Claude Code at the start. When the first scores were low, Claude Code suggested the models to try (mpnet, UniXcoder, e5-base-v2) and a longer input window; I approved each experiment.

Code generation or assistance: Claude Code wrote the repository's source code: model backends, query and snippet pre-processing, HyDE, multi-view embedding, BM25 hybrid scoring, versioned index, telemetry, demo CL. I directed the work, ran the code and reviewed the results.

Testing / debugging: Claude Code wrote and ran the 78 offline tests and diagnosed problems from the logs I sent back from real runs. I ran the real evaluations on my PC, Colab and Kaggle.

4. Feature Origin Classification (Please add/delete based on the number of the feature)

1. Feature Name: **F1: Retrieval pipeline and MTEB encoder (encoder, model backends, configuration)**
2. Self-Generated/AI-Generated/Both: **AI-Generated (directed, run and reviewed by me)**
3. Description (AI tools/platform used, prompt used, output summary, modification)

AI tool / platform used: Claude Code (Anthropic Claude models)

Prompt used: Attached the Build Plan PDF: “build to the best of your ability”.

Output summary: A Python package whose encoder plugs into MTEB (AbsEncoder), with a sentence-transformers backend, an offline hashing backend for tests, and one config object per experiment. It writes the results JSON in the format the problem statement shows.

Modification: Claude Code revised it after errors I reported from real runs

1. Feature Name: **F2: Model choice, e5-base-v2 with query/passage prefixes and a 512-token window (submitted: NDCG@10 0.1151, MRR@10 0.0986)**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description (AI tools/platform used, prompt used, output summary, modification)

AI tool / platform used: Claude Code proposed candidates, wrote configs and analysed results.

Prompt used: “so how do i achieve a better number ”

Output summary: Ten configurations measured on the real CoIR AppsRetrieval test split. e5-base-v2 scored best: +73.9% over the MiniLM baseline (0.0662).

Modification: I approved each experiment, had it run on Kaggle or Colab, and chose what to submit, adjusted the configs after each result.

1. Feature Name: **F3: Optional Retrieval Layers, Cross-Version Indexing, and Evaluation Infrastructure**

2. Self-Generated/AI-Generated/Both: **Both** (AI-assisted, with my contributions to validation, experimentation, and refinement).

3. Description (AI tools/platform used, prompt used, output summary, modification): Used Claude Code (Anthropic Claude models) with the Build Plan PDF and the prompt “build to the best of your ability.” The implementation covered retrieval enhancements (HyDE, multi-view, and hybrid search), cross-version indexing with content-addressed caching, and evaluation tools for ranking metrics, latency, and index costs. I reviewed the implementations, evaluated their performance against baselines, identified areas without measurable improvement, and refined the documentation, latency reporting, and model identification to ensure accurate results.

1. Feature Name: **F4: Tests, Dockerfile, Makefile**
2. Self-Generated/AI-Generated/Both: **Both(documentation written from results I supplied; reviewed by me)**
3. Description (AI tools/platform used, prompt used, output summary, modification)

AI tool / platform used: Claude Code (Anthropic Claude models)

Output summary: 78 offline tests (including the MTEB integration and the results-file format); a Docker image whose default run is the tests plus the demo; make targets for each step; setup and run instructions, the design, and a record of every experiment with its measured score.

Modification: Updated by Claude Code after each new result; claims later corrected to match the measurements.

1. Feature Name: **F5: Demo video (max 5 minutes)**
2. Self-Generated/AI-Generated/Both: **Both**
3. Description (AI tools/platform used, prompt used, output summary, modification)

AI tool / platform used: Claude Code (Anthropic Claude models) wrote the narration script and the list of what to show on screen.

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. **Yes**

No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: **Abhavya Jha**

Role: **Team Leader, Trace (solo participant)**

Signature: Abhavya Jha

Date: 30th September 2026